

ML-KEM.Decaps_internal(dk, c)

Uses the decapsulation key to produce a shared secret key from a ciphertext.

Input: decapsulation key $dk \in \mathbb{B}^{768k+96}$.

Input: ciphertext $c \in \mathbb{B}^{32(d_u k + d_v)}$.

Output: shared secret key $K \in \mathbb{B}^{32}$.

1: $dk_{PKE} \leftarrow dk[0 : 384k]$

▷ extract (from KEM decaps key) the PKE decryption key

2: $ek_{PKE} \leftarrow dk[384k : 768k + 32]$

▷ extract PKE encryption key

3: $h \leftarrow dk[768k + 32 : 768k + 64]$

▷ extract hash of PKE encryption key

4: $z \leftarrow dk[768k + 64 : 768k + 96]$

▷ extract implicit rejection value

5: $m' \leftarrow \text{K-PKE.Decrypt}(dk_{PKE}, c)$

▷ decrypt ciphertext

6: $(K', r') \leftarrow G(m' \| h)$

7: $\bar{K} \leftarrow J(z \| c)$

8: $c' \leftarrow \text{K-PKE.Encrypt}(ek_{PKE}, m', r')$

▷ re-encrypt using the derived randomness r'

9: **if** $c \neq c'$ **then**

▷ if ciphertexts do not match, “implicitly reject”

10: $K' \leftarrow \bar{K}$

11: **end if**

12: **return** K'